

SOME SENSORS AVAILABLE IN THE MARKET

Manufacturer	Sensor name/ Part number	Description	Features	Applications
Adeunis RF	LoRaWAN sensors	A ready-to-use radio transceiver allowing transformation of any type of 0V-10V, 4mA-20mA and I/O sensor into a wireless sensor	• Ready-to-use transceiver • LoRaWAN V1 network protocol • 0-10V, 4-20mA & I/O interfaces • Range up to 10km • Up to two sensors by transmitter • Up to ten years of autonomy on battery • IP67 enclosure	—
AMS	AS-MLV-P2	A gas sensor for monitoring indoor air quality	• Over-voltage, reverse-voltage protection • 2.7V for 320°C • 34mW at 320°C • Reflow capable • Available as tape on reel	Building automation and consumer air-quality monitoring applications
Bosch Sensortec	BMX055	A 9-axis sensor that allows accurate measurement of angular rate, acceleration and geomagnetic fields in three perpendicular axes	• Three sensors in one device, namely, accelerometer, gyroscope and magnetometer	• Advanced gaming, HMI and augmented reality • Advanced gesture recognition • Indoor navigation • Tilt measurement and compensation • Free-fall detection and drop detection for warranty logging • Display profile switching • Advanced system power management for mobile applications • Menu scrolling, tap/double-tap sensing
Exar	XR10910 - 16:1 sensor interface	A sensor-interface analogue front-end IC that integrates 16:1 differential multiplexer, offset-correction DAC, programmable-gain instrumentation amplifier and a voltage reference	• Force sensing • Integrated features for interfacing multiple bridge sensors with an MCU or FPGA: - 16:1 differential mux with I2C interface - Instrumentation amplifier - LDO - Offset correction DAC with I2C interface ($\pm 560\text{mV}$ offset correction range - RTI) • 1mV maximum input offset voltage • 100pA maximum input bias current • 556μA maximum supply current • 2.7V to 5V analogue supply voltage range • 1.8V to 5V digital supply voltage range • -40°C to +85°C temperature range • 6x6mm QFN-40	• Bridge-sensor interface • Pressure and temperature sensors • Strain-gauge amplifier • Weight scales • Industrial process controls
Greyhill	68P series sensors (68PARN-020C, 68PARS-020C)	Non-contact rotary position sensors that use Hall effect technology to produce ratiometric voltage output, which allows these to substitute for a potentiometer. With this output and contactless operation, these are ideal for use in applications that require constantly changing voltage control	• One million rotational cycles • Debris-resistant Hall effect sensor technology • Redundant Hall effect sensor • Ratiometric analogue output	• Agriculture vehicle control throttle • Test and measurement equipment • Audio/visual equipment • Industrial automation
PIL	P43	A ultrasonic sensor with digital output	• Range of 250mm to 6000mm • Two integrated internal LEDs for echo justification and analogue internal sensors • Three different output functions	• Object recognition for agriculture and fluid levels • Distance measuring for trucks and loading ramps • Level sensing and parking-lot detection/barrier monitoring
Texas Instruments	HDC1000	A digital humidity sensor with integrated temperature sensor; measures humidity based on a novel capacitive sensor	• Relative humidity operating range of 0%-100% • 14-bit measurement resolution • Relative humidity accuracy of $\pm 3\%$ • Temperature accuracy $\pm 0.2^\circ\text{C}$	• HVAC (heating, ventilating and air-conditioning) • Smart thermostats and room monitors • White goods • Printers • Handheld meters • Medical devices • Cargo shipping • Automotive windshield defog • Wearable devices • Mobile devices
Texas Instruments	TDC1000	A fully-integrated analogue front-end for ultrasonic sensing measurements of level, fluid identification/concentration, flow and proximity/distance applications common in automotive, industrial, medical and consumer markets	• Automotive AEC-Q100 (TDC1000-Q1) • Measurement range up to 8ms • Operating current of 1.8μA (2 SPS) • Transmitter channels TX1/TX2; supports single or dual transducer • Operating temperature range of -40°C to 125°C	• Measurements through tanks of varying materials: - Fluid level - Fluid identification/concentration • Flow metering: water, gas and heat • Distance/proximity sensing